THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 24, 2017 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg.°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. Which of the following methods is NOT suitable for hardness removal? (a) Lime-soda method, (b) nanofiltration, (c) recarbonation, (d) ion exchange, or (e) none of the above.
2. Which of the following air pollutants is a secondary pollutant? (a) CO, (b) CO_2, (c) NO, (d) SO_2, (e) ground level ozone, (f) stratospheric ozone, or (g) none of the above.
3. Which of the following statements is NOT correct with respect to alkalinity (ALK)? (a) ALK is the ability of a water to consume H^+ without significant pH change; (b) In natural systems, ALK is often dominated by HCO_3^-; (c) Adding CO_2 into a water does not change its ALK; (d) ALK is often reported in mg/L as $CaCO_3$; (e) ALK is increased during a typical coagulation process; or (f) none of the above.
4. Which process/devise is suitable for reducing NO_x emission from automobiles/vehicles? (a) Fuel cleaning, (b) three-way catalytic convertor, (c) baghouse, (d) selective catalytic reduction, (e) electrostatic precipitator, or (f) none of the above.
5. A typical microfiltration membrane is NOT able to remove: (a) TSS, (b) FSS, (c) TDS, (d) VSS, (e) sludge from activated sludge process, or (f) none of the above.
6. Which of the following statements is correct with respect to ocean acidification? (a) Ocean acidification is caused by global warming; (b) ocean acidification results in increased temperature of seawater; (c) ocean acidification results in reduced concentration of H^+; (d) ocean acidification results in reduced concentration of Ca^{2+}; or (e) none of the above.
7. Thermal stratification (a) only occurs in summer, (b) only occurs in spring, (c) is caused by over-saturation of oxygen, (d) is a direct consequence of eutrophication, (e) often results in oxygen depletion in eutrophic lakes, or (f) is closely related to global warming.
8. Which biological treatment is an anaerobic treatment? (a) Sludge digestion, (b) aerated lagoon, (c) trickling filter, (d) rotating biological contactor, (e) activated sludge, or (f) none of the above.
9. Which compound is a greenhouse compound but NOT an ozone depleting compound? (a) Halon-1211 (CF_2ClBr), (b) Halon-1301 (CF_3Br), (c) CFC-12 (CF_2Cl_2), (d) Methyl bromide (CH_3Br), (e) HCFC-22 (CF_2HCl), (f) methane (CH_4), or (g) none of the above.
10. Ammonia (NH_3) is a volatile compound that can easily partition into the gas phase. For this reason, it can be removed from water by air stripping. However, its protonated form (NH_4^+, $pK_a = 9.25$) is highly soluble in water. Which pH range gives the best nitrogen removal efficiency by air stripping? (a) $pH < 4.75$, (b) $4.75 < pH < 7$, (c) around pH 7, (d) $7 < pH < 9.25$, or (e) $9.25 < pH < 10$.

Section B (Three questions with unequal marks)

11. (a) A wastewater contains 200 mg/L $C_6H_8N_2$. Determine its nitrogenous oxygen demand and total theoretical oxygen demand.

(8 marks)

(b) Briefly explain COD, BOD_5, and ultimate BOD. Discuss their differences and explain how they are related to each other. How does the ratio of BOD_5/COD affect your choice of wastewater treatment technology?

(12 marks)

12. (a) A polluted air contains 20 ppmv CO. Determine the concentration of CO in mg/m^3. You can assume a temperature of 15 °C and an atmospheric pressure of 1 bar.

(6 marks)

(b) What is acid rain and how is it related to air pollution? Why does acid rain cause the erosion of building materials?

(12 marks)

(c) Explain the concepts of adiabatic lapse rate, ambient lapse rate, and air stability. How does the stability of air affect air pollution?

(12 marks)

13. (a) Disinfection is performed to a sand-filtered water by adding chlorine to a continuously stirred tank reactor. Based on the Chick Model (i.e., $dN/dt = - kCN$), perform a mass balance for the reactor and derive an expression for the disinfection efficiency.

(8 marks)

(b) The CT value required to achieve 1-log inactivation (90% removal of pathogen) is 35 mg·min/L. The flow rate of the sand-filtered water is 5000 m^3/d. The active chlorine concentration is 10 mg/L. Determine the required volume of the disinfection reactor in order to achieve a 4-log inactivation.

(6 marks)

(c) How to make the disinfection reactor more effective?

(6 marks)

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